The University of Zambia

in association with

ZCAS University

BFS322 TREASURY MANAGEMENT

SUPPLEMENTARY/DEFERRED EXAMINATION

25 January 2021

12:30 – 15:30 HOURS

TIME ALLOWED: **WRITING – THREE HOURS**

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer Three (3) questions from this section.
3. This question paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

QUESTION 1

Monetary Policy Committee Statement

Bank of Zambia Policy Rate Adjusted Upwards by 125 basis points to 11.50%

The Monetary Policy Committee (MPC) at its November 16-17, 2020 meeting decided to maintain the Policy Rate at 8.0% to moderate risks to financial stability and support growth, lives and livelihoods. This decision also allows monetary policy measures taken earlier in the year to take full effect. This is despite projections indicating that inflation will persist above the 6-8% target range throughout the forecast horizon. The Committee noted that financial stability remains fragile despite signs of marginal improvement in economic activity in the third quarter following the partial relaxation of COVID-19 restrictions. Although growth is expected to recover in the medium-term, limited fiscal space, as well as uncertainty surrounding the persistence of the COVID-19 pandemic and access to external financing remain key downside risks to growth prospects. In this regard, successfully navigating the debt restructuring process to restore debt sustainability and implementing fiscal and other structural reforms are critical to return to fiscal and macroeconomic stability.

(Extract from the November 2020, BOZ MPC statement.)

(i) Discuss the factors that the Bank of Zambia considers when arriving at the decision regarding the policy rate.

(ii) **(10 marks)**

(iii) The Financial Markets Department is in charge of the implementation of monetary policy at the Bank of Zambia. Critically analyse the role of this department in the implementation of monetary policy. **(20 marks)**

(iv) Discuss the Ministry of Finance treasury functions. **(10 marks)**

(Total: 40 marks)

SECTION B: **Attempt any THREE questions in this section**

QUESTION 2

Nkongole Enterprises Limited is a manufacturer of exotic furniture. You have been recently appointed as the company Treasurer. The company has been facing some working capital challenges recently, which has led it to rely on its overdraft facility with the bank on which it pays 20% p.a. in interest.

One of your first tasks as a Treasurer is to examine the management accounts of the company and you have been given the following information:

	2018	**2017**	**2016**
Cash	K0	K0	K5,000
Debtor days	90	60	30
Creditor days	120	100	45
Stock turnover per month	2	2.5	4

Required

i. Based on the above figures alone, advise management on the causes of the working capital problems that the company is facing. **(5 marks)**

ii. Explain why it is important for a firm to manage its working capital. **(5 marks)**

iii. One of the ways of managing working capital is to manage how much credit the company grants to its customers. Suggest ways in which the company can assess the creditworthiness of its customers. **(10 marks)**

QUESTION 3

You are the Treasury Assistant for a company. In your analysis of the company's bank accounts, you notice that the company maintains substantial credit balances in its current account. Upon enquiry, you are informed that it is the company's policy to maintain huge balances on the current account in order to reduce the risk of running out of cash.

(i) Explain to management why in your view the policy of maintaining high balances on the current account is not beneficial from a treasury point of view. **(4 marks)**

(ii) In order to manage its cash resources more prudently you have advised management to maintain a minimum balance of K1,000 on the current account. The bank pays an interest rate of 10% p.a. on deposits and charges K20 per transaction to transfer to or from the current account. From your experience the standard deviation of cash flows is K500 per day. Calculate the maximum balance that should be maintained on the current account and the return point at which transfers would be made into or out of the current account. **(8 marks)**

(iii) In order to prudently manage the surplus cash in the company's accounts, the treasurer will need to invest the surplus funds. What factors will the treasurer have to consider in deciding the appropriate investment options? **(8 marks)**

QUESTION 4

Herstatt risk is derived from the German Herstatt Bank which was closed in 1974 for insolvency. The operations of the bank were suspended in the middle of a foreign exchange transaction. Banks that had transferred currency to Herstatt Bank never received the agreed upon exchange currency. Therefore, the term Herstatt risk now refers to the risk of the transactional parties not settling the currency exchange.

(i) Describe two (2) types of settlement and two (2) types of transaction in the clearing and settlement process. **(4 marks)**

(ii) The Continuous Link Settlement (CLS) system was established in response to the settlement risk discussed above. Briefly discuss the operation of this system and how it helps in reducing settlement risk. **(4 marks)**

(iii) The Bank of Zambia has also put in place measures to try and improve the payments are settlement systems in Zambia. Write short notes on the following payment and settlement systems and for each system suggest any two advantages of the system.

(a) Zambia Interbank Payment and Settlement System (ZIPSS) also known as the Real-time Gross Settlement (RTGS) System. **(4 marks)**

(b) Electronic Funds Transfer (EFT). **(4 marks)**

(c) The Cheque Image Clearing (CIC) system. **(4 marks)**

QUESTION 5

The term structure of interest rates is the spread of interest rates that are paid on the same type of assets with different terms to maturity.

(i) Draw the three different types of yield curve and explain the reason for their shape.

(6 marks)

(ii) Discuss any two (2) theories that explain the factors that influence the term structure of interest rates. **(8 marks)**

(iii) A company has an outstanding loan of ZMK5 million (Kwacha Five Million) with the Bank on which it pays an interest rate of 15.2500% p.a (i.e. BOZ MPC Rate + 5%) The MPC rate is currently at 10.2500%. The loan is due for a reset in 3 months' time and the company is concerned that BOZ will increase the MPC rate. The company buys an FRA from the bank and locks in an interest rate of 16.5%.

What will be the net amount payable on the loan, should the MPC rate be adjusted to 12.0000% by BOZ in three months' time at the time of reset. Show all your calculations. **(6 marks)**

END OF EXAM

MARKING KEY

Question 1

(i) Factors that Bank of Zambia considers when arriving at MPC decision:

Inflation- The increase in the price of goods and services observed over a time period. Monetary authorities try to ensure a low inflation rate by targeting interest rates i.e. cost of credit in the country.

Economic growth- As measured by GDP i.e. the value of goods and services produced in the economy. The increase of interest rates will have an effect on GDP through the cost of credit that firms and individuals face in the economy.

Financial system stability- The banking sector plays an important role in the economy through the provision of credit to firms and individuals. The cost of credit will be affected by the decisions that monetary authorities make regarding monetary policy. High interest rates may lead to more delinquent loans and affect the stability of the financial system.

Exchange Rate- The monetary authorities may want to affect the rate of exchange between the domestic currency and foreign currencies by either increasing or reducing interest rates. The authorities seek a stable exchange rate in order to support economic growth.

(ii) Role of Financial Markets Department in implementation of monetary policy:

1. Implement monetary policy;

    - On 2nd April 2002 BOZ introduced the Policy Rate as the key interest rate to signal monetary policy stance
    - MPC meets every quarter to discuss policy rate
    - Adjustment to rate guided by economic indicators such as output, expected inflation and exchange rate
    - Bank has established corridor that defines the band within which overnight interbank rate is allowed to fluctuate
    - Bank's stance on whether to buy or sell financial assets determined by level of short term interest rates particularly in overnight market
    - Aim is to steer overnight interbank rate towards BOZ Policy Rate

2. Acting as fiscal agent of the government;

    - Executed through issuance and redemption of GRZ securities

- Conducts primary auctions of GRZ securities, redemptions and interest payments as well as maintaining GRZ securities databases
- Advises and provides domestic debt projections and estimations of domestic debt service to MOF for annual national budget purposes

3. Administering the Overnight Lending Facility;

- Department operates the OLF
- A liquidity financing facility which commercial banks can access in need of liquidity from BOZ at day end
- OLF complements other refinancing facilities such as TB Rediscount Facility and Intraday Credit Facility
- Primary purpose of OLF is to enhance operations of money markets by assuring commercial banks of liquidity assistance from BOZ

4. Managing foreign exchange reserves;

- Assets denominated in foreign currency held by monetary authorities
- Consist of cash, financial instruments, gold and SDRs
- International reserves held for the following reasons:
    - Intervening in the FX market
    - Execution of payments for exchange market
    - Servicing GRZ FX liabilities and debt
    - Providing funds during emergencies or national disasters
    - Support investor confidence that country can meet external commitments

5. Undertaking foreign exchange dealings;

(iii) Ministry of Finance treasury functions:

Role that MOF plays in managing the Country's treasury:

- Formulates and implements fiscal and monetary policy
- In charge of budget preparation and execution
- Manages the country's financial resources
- In charge of disbursing funds to the ministries for spending
- In charge of resource mobilisation e.g. through tax receipts
- In charge of public debt management

Question 2

(i) The working capital problems could be caused by the following: (1) for each point

- The company has run down its cash over the years
- This could be due to the fact that the cash conversion cycle has been lengthening over the years
- Its customers are taking longer to pay, as revealed by the worsening debtor days. This has the effect of tying up cash in the receivables
- It is taking the company longer to turn over its stock. This means that cash is increasingly tied up in unsold stock
- As a result of the worsening cash position of the company, it finds itself unable to meet its commitments as revealed by the worsening creditor days position.

(ii) The efficient management of working capital is important from the point of view of both liquidity and profitability. Poor management of working capital means that funds are unnecessarily tied up in idle assets, hence reducing liquidity and also reducing the ability to invest in productive assets such as plant and machinery, thereby affecting profitability.

(iii) Sources of information for credit standing of customers: (2 marks each)

- Bank references- most banks in Zambia offer references on their customers
- Trade references- suppliers already giving credit to the customer can give useful information about how good a customer is on baying bills
- Published information – customer's annual accounts and reports will give some idea of general financial position of company and its liquidity
- Credit agencies- agencies such as the Credit Reference Bureau in Zambia keeps information of individual and company borrowings from different banks

Company's own sales record- for an existing customer, the sales ledger will show how prompt a payer the company is

Question 3

(i) Holding large cash balances on the current account carries with it a cost i.e. the opportunity cost of the profits which could be made if the cash was used in the company or invested the cash elsewhere.

(ii) Spread = $3 \times \left(\frac{3 \times \text{transaction cost} \times \text{variance}}{4 \quad \text{interest rate}}\right)^{1/3}$

$$= 3 \times \left(\frac{3}{4} \times \frac{K20 \times K250{,}000}{0.00027}\right)^{1/3}$$

= 3 x 2403.75

= 7, 211.25

Upper limit = lower limit + spread

= K1,000 + K7, 211.25

= K8, 211.25

Return point = lower limit + 1/3 x spread

= K1,000 + 1/3 x K7,211.25

= 3, 403.75

Factors to consider in deciding the appropriate investment options:

- Length of time for which funds are available;
- Amount of funds available;
- Return offered on investment in relation to amount;
- Risks associated with calling in investment early;
- Ease of realization

Question 4

(i) Settlement types

Gross basis

- where each bank pays the total amount owed.
- Payments handled through a gross basis are more likely to have a requirement for immediate execution, where payment instructions are processed separately for each individual transaction

Net basis

- where a large number of transactions are accumulated and offset against each other, with only the net difference being transferred between banks
- Payments usually wait until the end of the day, when all transactions between the banks are summarized and offset against each other

Transaction types:

High-value payments

- must be executed immediately
- Because of the time constraints, high value payments cannot wait for end-of-day netting and so are settled on a gross basis, with an immediate transfer between the banks

Low-value payments

- does not require immediate execution and tends to be for smaller amounts.
- These payments are handled through a net settlement system

(ii) Continuous Link Settlement (CLS)

- This is called "settlement risk" or "Herstatt risk" after the German Bank that failed.
- To avoid this settlement risk, while also speeding up the settlement process, the CLS system was created by a number of banks
- This system is operated by CLS Bank International, of which the founding banks are shareholders

- Other banks can submit their foreign exchange transactions through these member banks.
- Only a select number of foreign currencies are transacted in the CLS system

(iii) Payment systems in Zambia:

(a) Real Time Gross Settlement System (RTGS)

- ZIPSS is an electronic payment system in which processing of transactions for settlement takes place continuously on a transaction basis in real time
- ZIPSS was developed to improve the management of payment risks related to high value payments.
- It provides a more secure platform for making both large value payments
- It also provides quicker final and irrevocable payments

(b) **Electronic Funds Transfer (EFT)**

- The Electronic Funds Transfer (EFT) system is a simple, safe and speedy way to make and collect payments by electronic transfer
- Safe and secure transfer of funds from one account to another
- Payment of wages, salaries and pensions
- Payment of utility bills
- Disbursement of loans and collection interest payments by financial institutions

(c) The Cheque Image Clearing System (CIC)

- The Cheque Image Clearing (CIC) system is a cheque processing and clearing system and allows customers belonging to one bank to write cheques belonging customers of another bank.
- Similar to Electronic Funds Transfer (EFT) system, the CIC system is operated by the Zambia Electronic Clearing House Ltd (ZECHL).
- With the implementation of the truncation system in 2014, substantial improvement in turn-around time in cheque processing have been achievement.

- Credit to customer accounts is given on a T+1 basis.

Question 5

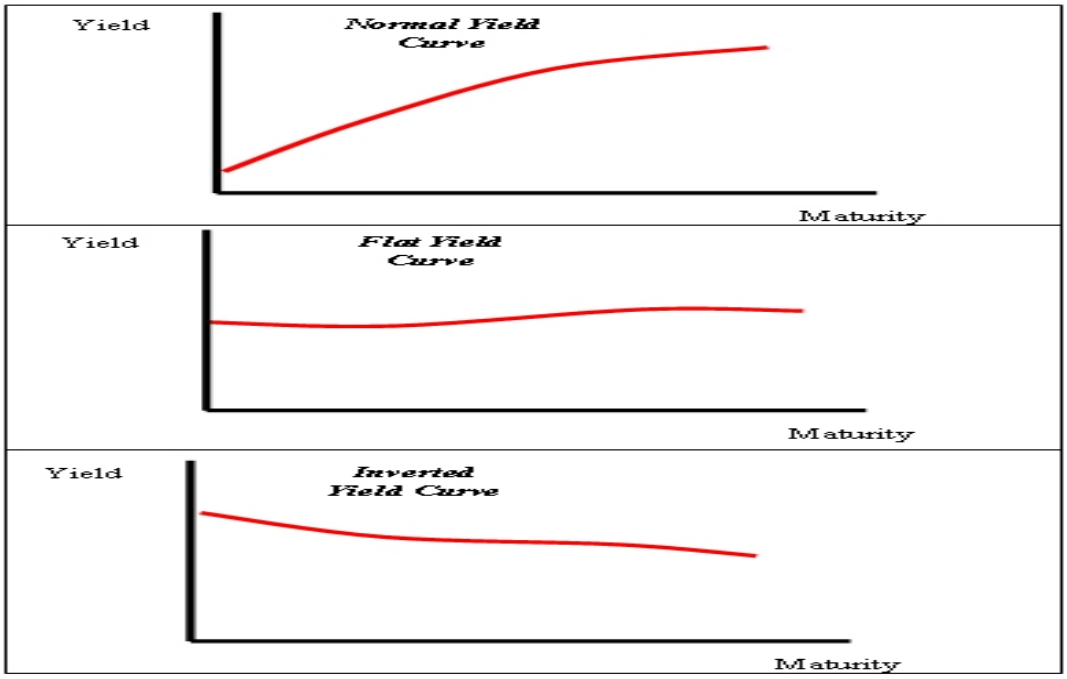

Figure 1

1. Normal Yield Curve

- A normal yield curve slopes upwards as borrowers who wish to borrow for long periods of time pay a higher rate of interest than short term borrowers
- The characteristic upward curve is attributable to liquidity preference and to risk
- The overall risks associated with lending increase over time, because other risks increase

2. Inverted Yield Curve

- An inverted yield curve slopes downwards and indicates that borrowers who wish to borrow in the short term pay a higher rate of interest than long term borrowers.
- An inverted yield curve is a sign of an impending recession as investors prefer to lend short due to the uncertainties of lending long in a depressed economy.

3. Flat yield Curve

- A flat yield curve shows that short-term borrowers and long term borrowers pay almost the same rate.
- It characteristic of an very stable economy where the risk factors are low.

(i) Interest Rate Theories:

1. Expectations theory
    - States that the forward interest rate is due solely to expectations of future interest rate movements.

2. Liquidity Preference Theory

    - Bond holders are generally risk-averse and will wish to be compensated for the higher risk involved in holding long-term bonds by an appropriate risk premium
    - According to the liquidity preference theory, the yield on longer-term bonds will therefore reflect not only market expectations but also a liquidity premium

3. Market Segmentation theory

    - Assumes that investors have preferred habitats for their investments
    - Assumes that there are barriers to switching between short, medium and long term investments
    - According to the theory, the shape of the yield curve is determined by separate supply and demand forces in each particular segment

4. Preferred habitat theory

    - Theory argues that the term structure reflects both the expectations of the future path of interest rates and a liquidity premium
    - According to the theory, investors have specific objectives for their investments and this usually leads to specific time horizons (preferred habitats)

(ii) Loan Amount = K5,000,000
FRA Rate = 16.5%
Rate in 3 months' time = 12% + 5% = 17%
FRA compensation = 17.0% - 16.5% = 0.5%

Interest payable on loan (17% x K5m x 1)	= K850,000
Amount receivable on FRA (0.55 x K5m x 1)	= (K25,000)
Net amount payable by company	= K825,000